TELEPROMPTER SCRIPT & VIDEO RECORDING (AUDITED VERSION)

Presenter: Thiago da Silva
Project: Credit Performance Diagnostic and Scale Reassessment (Happen Bank / Lending Club)
Target Time: ~11:00 to 12:00 minutes

SLIDE 01: EXECUTIVE COVER & ROBUSTNESS

[CAMERA ON SCREEN / CAMEO ACTIVE — EYES FIXED ON LENS]
[TONE: DEEP, CONFIDENT, HIGH AUTHORITY]

"Hello everyone. I'm Thiago da Silva, and today I'm presenting the analytical diagnosis of credit performance for the LendingClub portfolio, currently HappenBank, available on Kaggle.
I'd like to start by highlighting that we're looking at a highly mature, solid, and winning operation. Between 2007 and 2018, we originated **$20.3 billion** in fixed-rate personal credit, spread across **1.34 million contracts** fully closed out, generating accumulated net profit of **$3.33 billion**.

[SLOW DOWN SLIGHTLY — HOLD EYES ON LENS]

One important point about the method: the nearly 11 years of origination analyzed were inflation-adjusted to Constant 2018 Dollars, and span from the 2008 crisis through the following expansion. In other words, the numbers ahead aren't a snapshot of a single year — they're a reliable average across almost eleven years.
Even at this scale, the data proves that Happen Bank already manages risk and balances its portfolio exceptionally well. Our historical ROA of **4.43% per year** beats our internal target of 4.00% and the market's High-Yield competitors, like Synchrony at 3.80% and Discover at 2.70%.

[PAUSE — HOLD EYES ON LENS]

So the goal of this investigation isn't to fix a bank with problems — it's to fine-tune an operation that already works well, mapping up to **$299.5 million** in accumulated opportunity cost. The reason is simple: in a multi-billion-dollar operation, small pricing gaps compound into large numbers."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 02: THE BUSINESS MODEL & THE ROLE OF INVESTORS

[TONE: FLOWING AND EXPLANATORY]

"In practice, Happen Bank works as a digital credit platform. We do everything 100% online, directly connecting people who need money with people who want to invest, without physical branches or the heavy structure of a traditional bank.

Our main product is the unsecured personal loan, with a fixed term of 36 or 60 months, used by most customers to pay off expensive credit card debt. Each application also receives its own risk grade, from A to G, which sets the interest rate according to each customer's chance of default — different from the income Class we'll see next, and central to the analyses ahead.

On this platform, the key piece is the investor: it connects borrowers to investors, both individuals and institutions, who receive back the interest paid by customers. That's why the sustainability of our model depends 100% on delivering an attractive, predictable net return to whoever puts up the capital."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 03: THE CENTRAL METRIC: THE CONCEPT OF ROA IN CREDIT

[TONE: ANALYTICAL AND ASSERTIVE]

"To measure whether we're compensating that capital adequately, we use the portfolio's ROA metric. It represents accumulated net financial result — already netting out real default losses and operating expenses — divided by the total amount lent.

The requirement for an internal target, a minimum return of **4.00% per year** of ROA, is explained by the absence of real collateral: in case of default, we don't have a house or car to repossess, which makes recovery depend purely on collections. That's why our pricing needs to build in a robust safety margin against risk.

And as we've seen, our track record of **4.43% ROA** confirms the bank already masters the balance between volume and risk. Across 1.34 million contracts and $20.3 billion originated, data analysis lets us pinpoint exactly where our scale is hiding profit margin."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 04: ANALYTICAL ARCHITECTURE: SEGMENTATION

[TONE: TECHNICAL AND STRUCTURED]

"To make this diagnostic possible, I built two proprietary segmentation variables, using SQL, that let us see past the bank's averages.

The first is the Income Bracket, built directly from the data: we split the portfolio into five similarly sized groups, with about 268,000 contracts each — from Class A, with income above $104,000, to Class E, the most vulnerable, with income up to $44,100.

The second variable measures how much this loan's installment weighs on the customer's budget: the market usually looks at total debt (DTI), treating anything above 30% as high;

here, when this installment alone consumes more than 20% of monthly income, the data shows risk turns critical — that's the profile we call Choked."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 05: THESIS 1 — THE SILENT DANGER OF THE "MIDDLE"

[TONE SHIFT — VALUE THE UNDERWRITING FIRST, THEN EMPHASIZE THE MIDDLE'S LOSS]

"The most important discovery in this analysis was debunking a myth about the Choked customer. The risk committee's traditional intuition tends to focus on the borrower with more than 20% income commitment, since that's where ROA plunges to 1.37%.
That group isn't actually our biggest problem, because our underwriting pipeline already filters these cases well before the contract closes. This profile represents just **13,163 contracts**, or 0.98% of our base, with **$319 million financed** and **$27.6 million** in opportunity cost. This shows our entry filter is already working.

[SLOW DOWN AND EMPHASIZE]

The real cash drain is in the Middle, between 10% and 20% commitment: **$7.40 billion** across **367,023 contracts**, 27.29% of all the bank's sales. Because of scale — 28 times larger than the Choked bracket — small pricing gaps generated **$239.2 million** in opportunity cost: **80% of the bank's entire loss** relative to our minimum return target."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDES 06 & 07: LOSS EPICENTERS AND INTEGRATED MATRIX

[TONE: SURGICAL AND DATA-DRIVEN]

"Looking more closely at the numbers in the Integrated Matrix on screen, most of the $239.2 million in opportunity cost in the Middle is concentrated in Grades C, D, and E, within the 10% to 20% income-commitment band.
For example, Grade D in the Middle moved 74,151 contracts and **$1.50 billion**, but ran at an ROA of just 2.76% — well below our 4.00% floor — generating an opportunity cost of **$78.6 million**.
Adding up Grades C, D, and E, we get **222,055 contracts**, **$4.48 billion financed** and **$185.7 million** in opportunity cost — 78% of the Middle's relative loss. In other words, we generated a large volume, but charged less than we should have for the real risk of that scale."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDES 08 & 09: THESIS 2 — TERM ELASTICITY & BREAKDOWN

[TONE: CLEAR CONTRAST — POSITIVE TONE FOR CLASSES A/B, SOBER TONE FOR CLASSES D/E]

"Our second thesis investigates the impact of loan term and grounds the concept of Term Fatigue. Looking at the bank's consolidated average, extending to 60 months seems advantageous, but that average hides a trap: opposite behaviors across income brackets.

[POSITIVE TONE]

You can see this in Classes A and B, the high-income brackets, where extending to 60 months acts as a value lever. In Class A, with 268,487 contracts and $4.21 billion financed, ROA rises from 5.29% at 36 months to 6.24% at 60 months, a gain of **+0.95 p.p.** that generates **+$96.0 million** in nominal gain. In Class B, with 269,294 contracts and $3.95 billion, ROA climbs from 4.59% to 4.95%, growing **+0.36 p.p.** and adding **+$32.2 million**. Together, they capture **$128.2 million**. Class C, meanwhile, stays neutral, with a marginal impact of -$11.9 million.

[SOBER, ATTENTIVE TONE]

On the other hand, for the lower-income Classes D and E, the 60-month term turns into a trap: vulnerable families find it increasingly hard to pay starting around month 36. As a result, in Class D, with 268,295 contracts and $3.41 billion, ROA falls from 3.94% to 3.00%, shrinking **-0.94 p.p.** and destroying **-$51.4 million**. In Class E, with 271,315 contracts and $3.15 billion, ROA plunges from 2.91% to 1.58%, dropping **-1.33 p.p.** and racking up a nominal loss of **-$34.2 million**. Combined, the accumulated loss across these two classes reaches **$85.6 million**. Across all five classes combined, the net effect is just +$30.7 million — and it's this net number, not the extremes, that should guide the committee's decision."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 10: SPECIAL CASE STUDY — THE GRADE G PARADOX

[TONE: REVEALING AND EXPLANATORY]

"To wrap up the diagnostic, the analysis of Grade G — our worst credit grade, with **9,126 contracts** and **$196.4 million financed** — presents an unusual and highly relevant case of risk-based pricing.
In this group, the pattern is the reverse of the income brackets: at the short 36-month term, Grade G runs in the red, with an ROA of **-1.52%** across 1,518 contracts and $23.1 million financed — a loss of **-$1.05 million**. At the 60-month term, though, it recovers: an ROA of **+3.76%** across 7,608 contracts and $173.3 million financed, delivering **+$32.59 million** in real net profit.
This gap comes from the interest-accrual window: at 36 months, the term is too short for interest income to offset the early, concentrated defaults; at 60 months, even with more defaults in absolute terms, the margin charged to paying customers over 5 years is enough

to cover the losses and still return **$32.59 million** in positive net result. This proves the problem was never approving Grade G — it was offering it the wrong term at 36 months."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 11: STRATEGIC ACTION PLAN

[TONE: EXECUTION, DECISIVE, C-LEVEL RECOMMENDATION]
"Given these findings, I'm presenting three practical, actionable recommendations for the committee, focused on tuning our origination engine:
First, the **Surgical Term Lock**, which blocks the 60-month option for Classes D and E, routing 100% of originations to the 36-month term — avoiding up to **$85.6 million** in term-fatigue losses.
Second, **Repricing the Middle**, applying a rate adjustment of **+45 to +80 basis points** on Grades C, D, and E within the 10% to 20% income-commitment band, directly recovering up to **$123.75 million** in margin.
Third, **Reallocating Grade G**, ending the 36-month offer for this risk grade and channeling 100% of applications to the 60-month term — eliminating the loss of **-$1.05 million** from the short term and adding **+$4.3 million** in incremental gain. It's worth noting these three levers don't add up: the Surgical Term Lock and Reallocating Grade G correct term length, based on our Thesis 2 elasticity analysis; only Repricing the Middle corrects rate and ties to the $299.5 million diagnosed in this portfolio — $123.75 million already actioned in this phase, with $175.75 million moving to the next stage, across Grades A/B, F/G, and the Choked segment."

[TECHNICAL PAUSE, 1 SECOND — ADVANCE SLIDE]

SLIDE 12: CONCLUSION & UNLOCKED VALUE PROJECTION

[TONE: CONFIDENT, CONCLUSIVE, ENERGETICALLY ALIGNED]

"In conclusion, this data shows that the path to maximizing Happen Bank's value doesn't require cutting our origination — it requires precisely tuning the term and price of the Middle of the table.
The most measurable gain comes from Repricing the Middle: the projection points to a rise in ROA from **4.43% to 4.57% per year**, recovering in this first phase **$123.75 million** in margin, already consolidating our position at the top of the High-Yield market. The Surgical Term Lock and Grade G reallocation add to this result on their own separate basis — avoiding $85.6 million and unlocking a further $4.3 million, with no overlap against the $123.75 million already counted. Of the $299.5 million diagnosed, the remaining $175.75 million moves to the second phase of sub-population repricing.
It's worth noting that the SQL data engineering is complete, the models are validated, and the Power BI dashboards are ready to track execution by vintage.

[HOLD EYES STEADY ON LENS — LIGHT, PROFESSIONAL SMILE]

Thank you very much for your attention."